# Modulating the Second Peak of a First-Passage-Time Distribution by Scanning the Destination of a Directed Shortcut (Fixed N=100, K=6; Analytic inversion, flux validation, and Monte Carlo)

## 1 Introduction and objective

Under the start/target configuration used in the main manuscript, I fix $(N, K) = (100, 6)$ and vary only the shortcut destination dst. I aim to:

- quantify how the peak timings and the relative second-peak prominence (e.g., $R = A(t_2)/A(t_1)$) change with dst;
- estimate, among trajectories whose absorption times fall near the second peak, the conditional shortcut-crossing probabilities.

## 2 Setup

### 2.1 Model and fixed parameters

- Nodes are labeled $1, \ldots, N$ with $N = 100$.
- Initial node: $n_0 = 1$.
- Absorbing target: target $= N/2 = 50$ (strict first passage; $\rho = 1$).
- Ring connectivity: $K = 6$ neighbors $(\pm 1, \pm 2, \pm 3)$.
- A single directed shortcut from src to dst, where dst is scanned.
- I compare two choices of shortcut source:
  - Case A: src $= n_0 + 5 = 6$.
  - Case B: src $= n_0 = 1$.

## 3 Main conclusions

- Scanning only dst (with $N = 100$, $K = 6$, $n_0 = 1$, target $= 50$ fixed) strongly modulates the peak timings and the *relative* prominence of the second peak (captured by the ratio $R = A(t_2)/A(t_1)$). In the late-second-peak family, the absolute second-peak height $A(t_2)$ varies only weakly across dst, except for special near-target cases.
- A deterministic flux/master-equation scan reproduces the bimodal dst sets and matches the AW inversion over $t \leq 2000$ (max absolute error $\sim 10^{-13}$ on representative cases), indicating the bimodality is not a numerical artifact.
- For late-second-peak cases, the conditional shortcut-usage probability is more cleanly organized by the peak ratio $R$ (excluding the special dst = target point): empirically $P(C \geq 1 \mid T \in W)$ tends to decrease as $R$ increases. $A(t_2)$ shows a positive association with $P(C \geq 1 \mid T \in W)$, but its dynamic range is comparatively small in the late family.
- Case B (src $= 1$) additionally exhibits an early-$t_2$ family when dst is close to the target, due to a two-step absorbing path; in these cases, conditional on $T \in W$ shortcut usage is typically near-deterministic ($P(C \geq 1 \mid T \in W) \approx 1$).

- If I focus on Case A, the first/second peaks are typically smoother late-time features (as opposed to the near-delta early peaks in Case B), and the conditional crossing fractions are correspondingly more informative and stable across dst (with visibly nontrivial mass at $C = 0$, $C = 1$, and sometimes $C \geq 2$).

# 4 Methods

## 4.1 Second-peak identification and MC conditioning

- I use the peak-selection rule from the caption of Fig. 3 in the manuscript: local maxima with $A(t) > 10^{-7}$; the second peak must reach at least 1% of the global maximum.
- $A(t)$ is a discrete-time first-passage-time probability mass function, so the plotted polyline is not expected to be smooth.
- Denote the first two peaks as $(t_1, t_2)$. I define a symmetric window around the second peak via

$$\Delta = \max\{1, \lfloor 0.05\,(t_2 - t_1)\rfloor\}, \qquad W = [t_2 - \Delta,\ t_2 + \Delta].$$

  In MC I retain only trajectories with absorption time $T \in W$ and record their shortcut crossing count $C$ (0, 1, or $\geq 2$).

## 4.2 Estimands and Monte Carlo estimators

Each trajectory yields a pair $(T, C)$:

- $T$: first-passage absorption time (the number of steps until first hitting target).
- $C$: the shortcut traversal count, i.e., the number of times the edge src $\to$ dst is used.

The main conditional quantities of interest are

$$P(C = 0 \mid T \in W), \qquad P(C \geq 1 \mid T \in W) = 1 - P(C = 0 \mid T \in W),$$

and additionally $P(C = 1 \mid T \in W)$ and $P(C \geq 2 \mid T \in W)$. If $n = |\{i : T_i \in W\}|$ and $k_0 = |\{i : T_i \in W,\ C_i = 0\}|$, then

$$\widehat{P}(C = 0 \mid T \in W) = k_0/n, \qquad \widehat{P}(C \geq 1 \mid T \in W) = 1 - k_0/n.$$

For uncertainty visualization I use Wilson 95% confidence intervals for the Bernoulli event $\{C \geq 1\}$ (computed by the plotting script). I also report the window hit rate $n/N_{\text{walkers}}$; if $n$ is too small (e.g., $< 10^3$), the estimated fractions have substantial sampling error.

# 5 Deterministic validation (flux/master equation)

## 5.1 Method (time-domain flux / master equation)

Treat target as absorbing, propagate the transient mass $p_t(i)$ deterministically, and read off the FPT pmf as the flux into the target:

$$f(t+1) = p_{t+1}(\text{target}), \qquad p_{t+1}(j) = \sum_i p_t(i) P(i \to j),\ j \neq \text{target}.$$

Implementation uses a directed edge list $(\text{src}, \text{dst}, w)$ with $w = 1/\deg(\text{src})$ for a sparse $O(E)$ update per step (no dense $(N \times N)$ matrix). This provides a deterministic check that bimodality is not a Monte Carlo artifact and that the AW inversion agrees with a time-domain baseline.

### 5.2 Bimodality criteria (Fig. 3 + separation)

- Fig. 3-caption style: strict local maxima with $f(t) > 10^{-7}$; the second peak must be at least 1% of the global maximum peak height.
- Separation/robust filter: additionally require the two peaks to be separated by at least $2K = 12$ to exclude early "micro-peak clusters".
- Valley-depth diagnostic: with $h_{\min} = \min\{f(t_1), f(t_2)\}$,

$$V = \frac{\min_{t_1<t<t_2} f(t)}{h_{\min}}.$$

  $V \approx 1$ means the valley between the two peaks is shallow.

### 5.3 Scan results (Case A: src = 6)

With $(N, K, n_0, \text{target}) = (100, 6, 1, 50)$ fixed and scanning dst:

- Fig. 3 criterion finds 15 bimodal destinations dst $\in \{43, \ldots, 57\}$ (including the special dst = target = 50).
- The separation/robust filter keeps 14: dst $\in \{43, \ldots, 49, 51, \ldots, 57\}$.

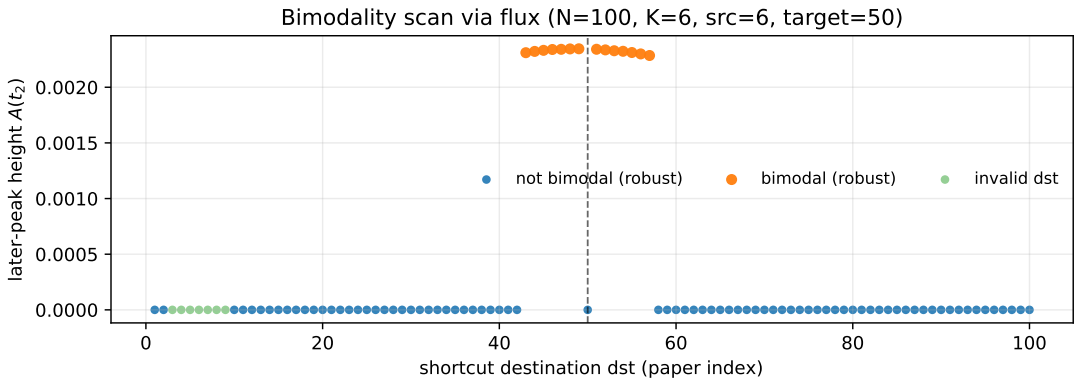


Figure 1: Case A: flux/master-equation scan over dst and the late-peak height $A(t_2) = f(t_2)$ (dashed line: dst = target = 50).

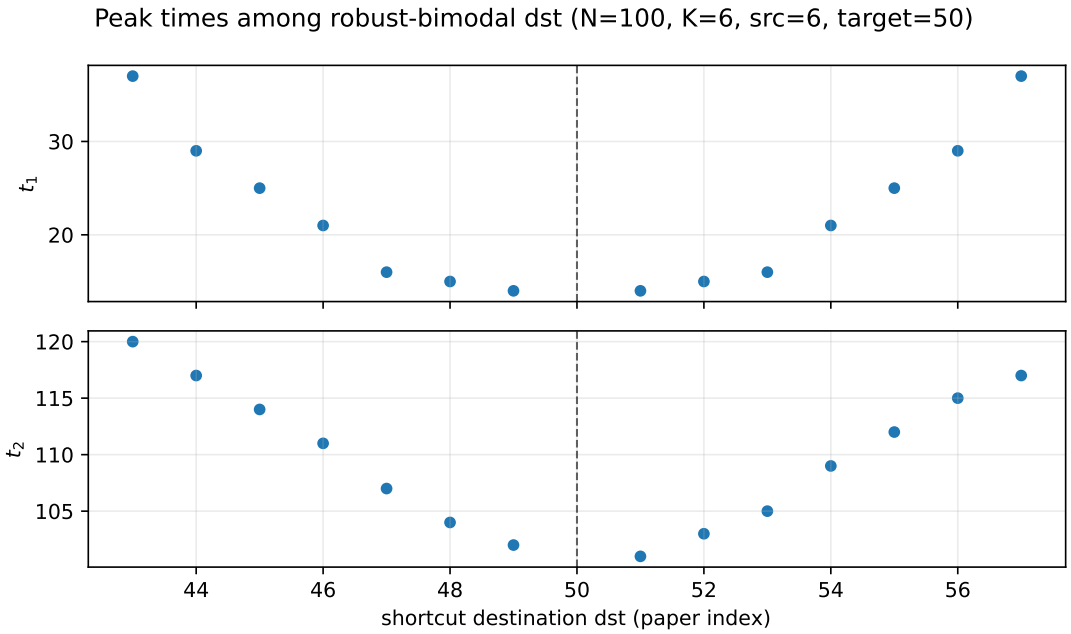


Figure 2: Case A: peak times $t_1, t_2$ among robust bimodal destinations.

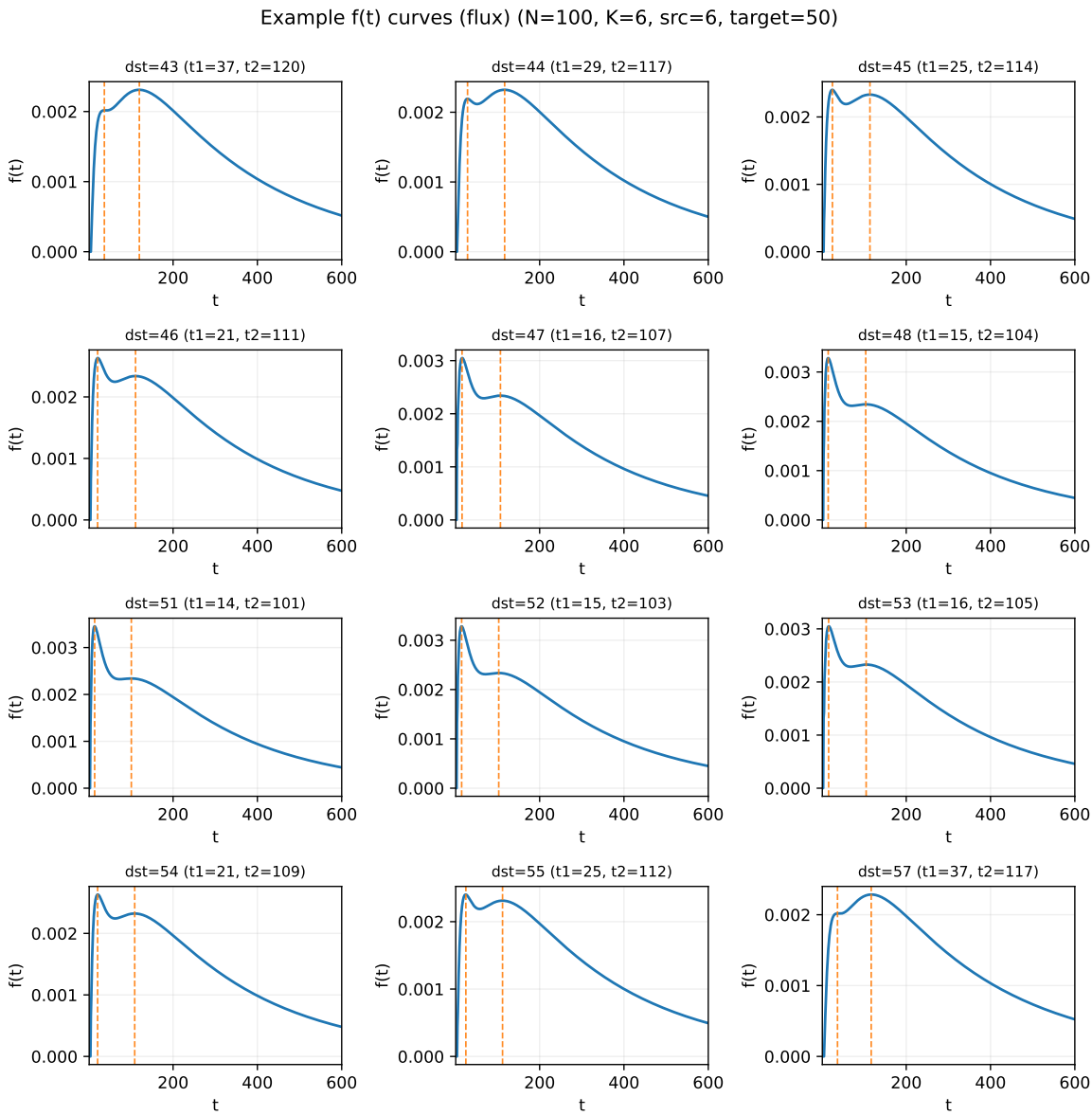


Figure 3: Case A: representative robust bimodal pmfs $f(t)$ (flux), with $t_1, t_2$ marked.

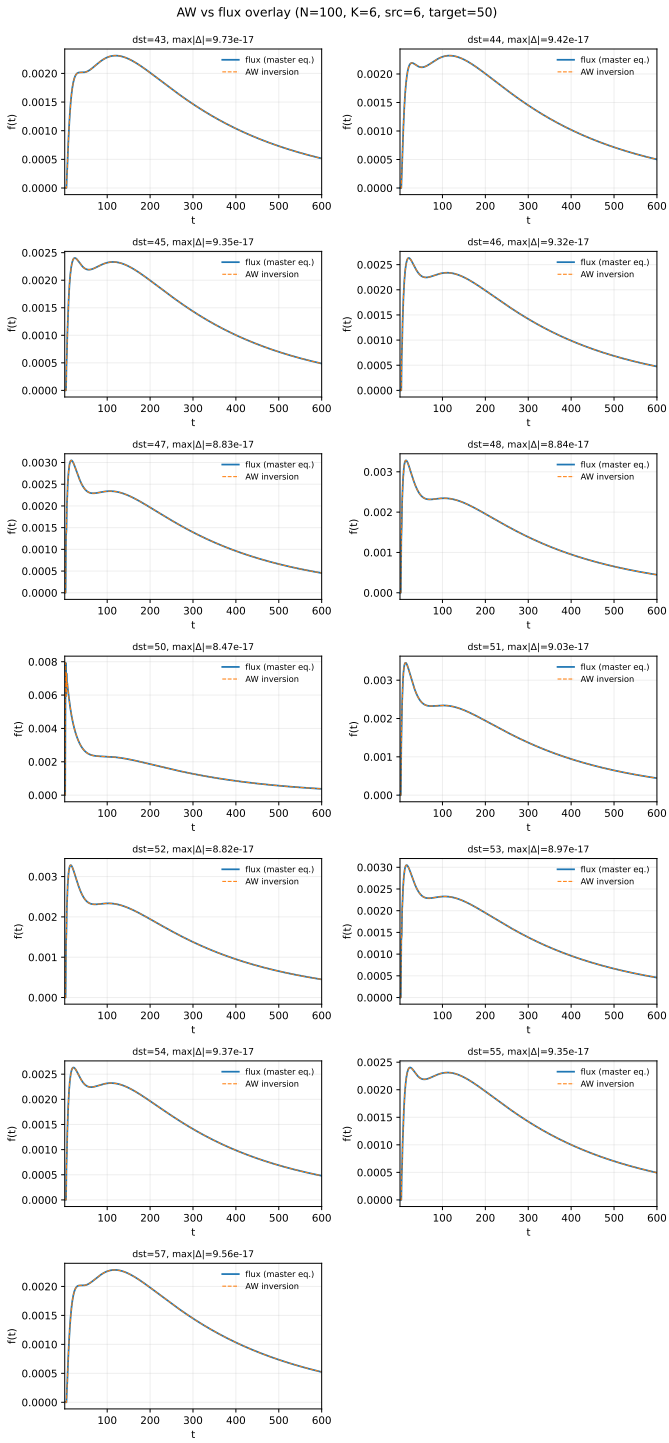


Figure 4: Case A: AW inversion vs. flux $f(t)$ overlays (max error $\sim 10^{-13}$).

| dst | $t_1$ | $t_2$ | $t_2 - t_1$ | $A(t_1)$ | $A(t_2)$ | $V$ |
|---|---|---|---|---|---|---|
| 43 | 37 | 120 | 83 | 0.00201691 | 0.00231041 | 1.000 |
| 44 | 29 | 117 | 88 | 0.00219366 | 0.00232199 | 0.965 |
| 45 | 25 | 114 | 89 | 0.00240315 | 0.00233158 | 0.940 |
| 46 | 21 | 111 | 90 | 0.00263207 | 0.00233914 | 0.960 |
| 47 | 16 | 107 | 91 | 0.00304909 | 0.00234085 | 0.979 |
| 48 | 15 | 104 | 89 | 0.00328311 | 0.00234431 | 0.987 |
| 49 | 14 | 102 | 88 | 0.00345483 | 0.00234556 | 0.992 |
| 51 | 14 | 101 | 87 | 0.00345483 | 0.0023412 | 0.993 |
| 52 | 15 | 103 | 88 | 0.00328311 | 0.00233559 | 0.989 |
| 53 | 16 | 105 | 89 | 0.00304909 | 0.00232787 | 0.982 |
| 54 | 21 | 109 | 88 | 0.00263207 | 0.00232303 | 0.965 |
| 55 | 25 | 112 | 87 | 0.00240315 | 0.00231205 | 0.947 |
| 56 | 29 | 115 | 86 | 0.00219365 | 0.00229944 | 0.965 |
| 57 | 37 | 117 | 80 | 0.0020168 | 0.00228525 | 0.999 |

### 5.4 Scan results (Case B: src = 1)

- Fig. 3 criterion finds 18 bimodal destinations, but many are early-time peak clusters (separation $< 2K$).
- After the separation/robust filter, 7 remain as typical "late second peak" cases: dst $\in \{36, 37, 38, 39, 62, 63, 64\}$.

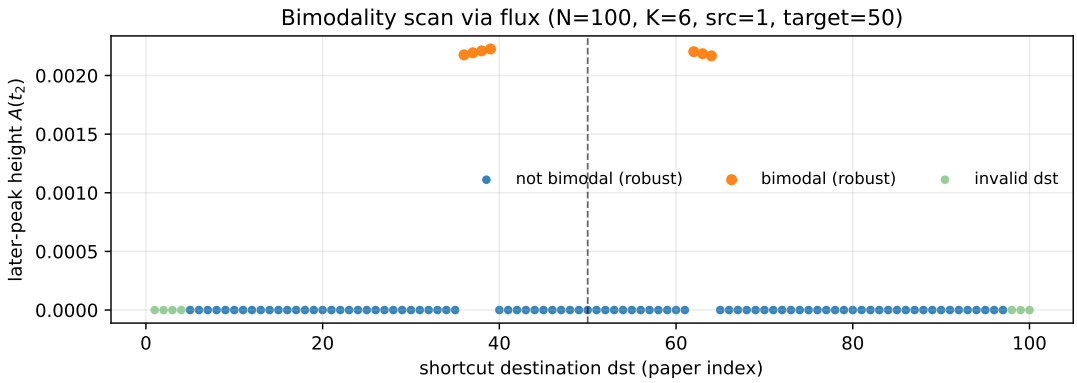


Figure 5: Case B: flux scan over dst and the late-peak height $A(t_2) = f(t_2)$ (robust cases highlighted).

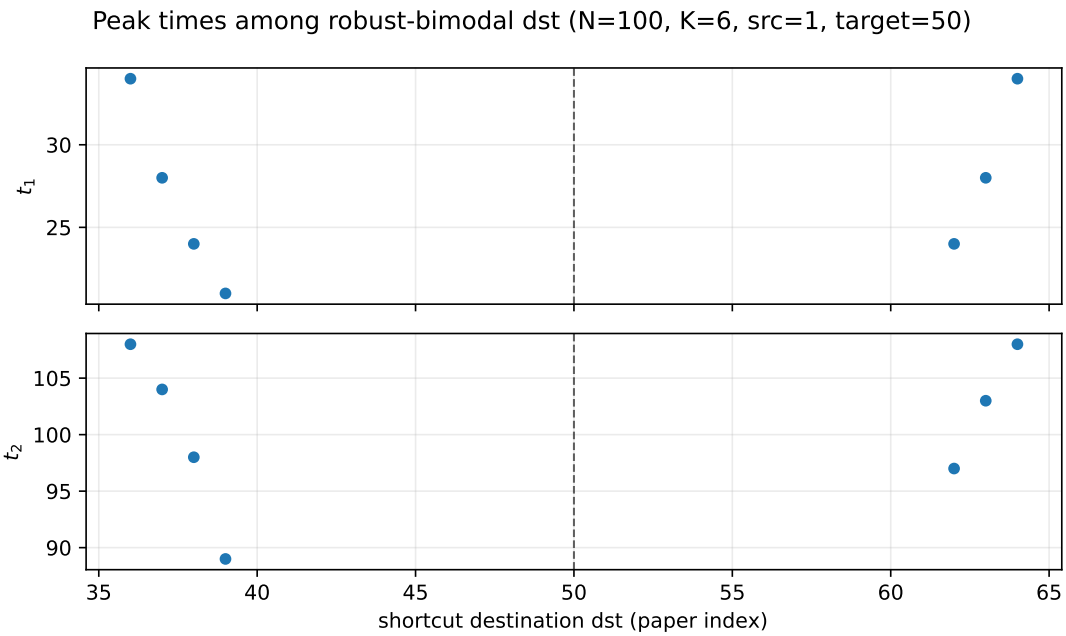


Figure 6: Case B: peak times $t_1, t_2$ among robust bimodal destinations.

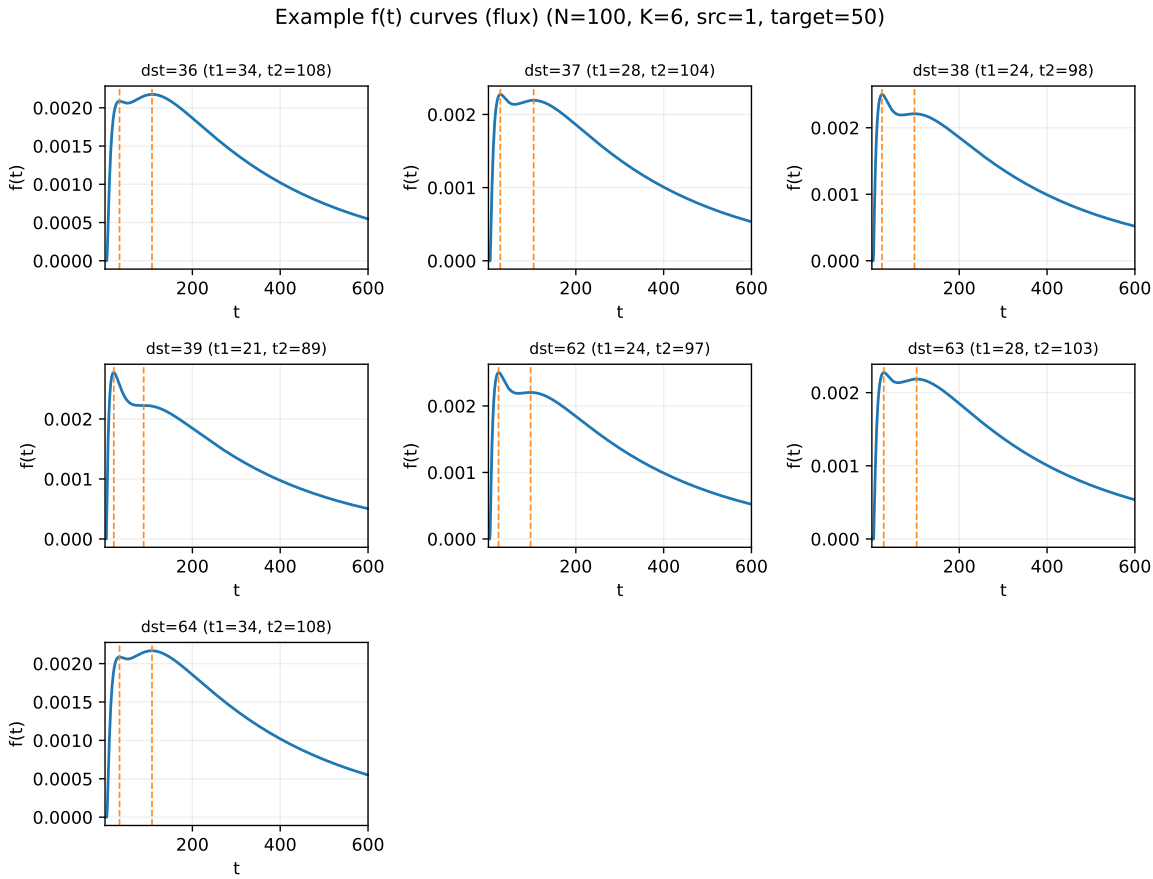


Figure 7: Case B: representative robust bimodal pmfs $f(t)$ (flux), with $t_1, t_2$ marked.

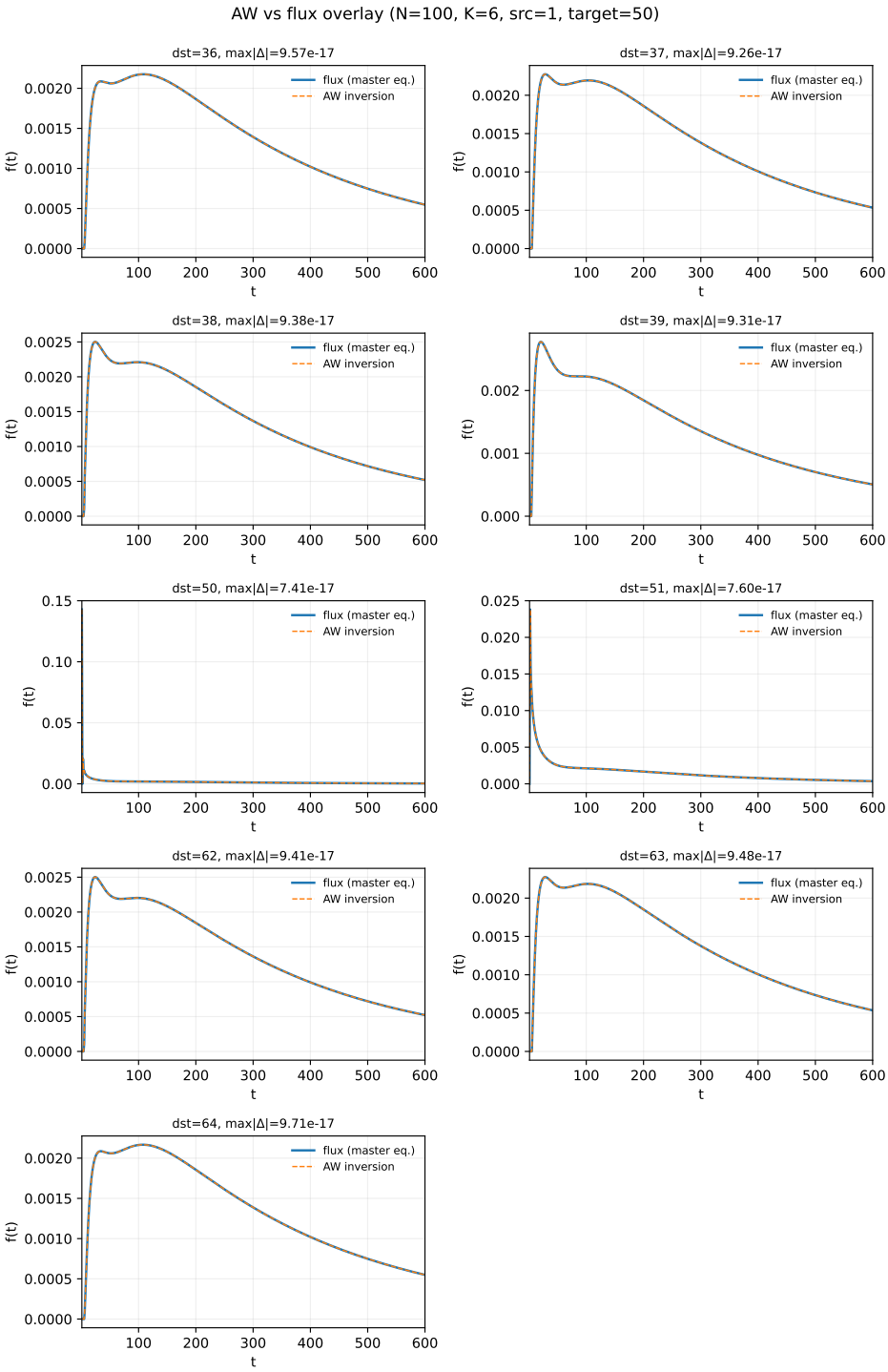


Figure 8: Case B: AW inversion vs. flux $f(t)$ overlays (max error $\sim 10^{-13}$).

| dst | $t_1$ | $t_2$ | $t_2 - t_1$ | $A(t_1)$ | $A(t_2)$ | $V$ |
|---|---|---|---|---|---|---|
| 36 | 34 | 108 | 74 | 0.00208676 | 0.00217576 | 0.988 |
| 37 | 28 | 104 | 76 | 0.00227434 | 0.0021934 | 0.975 |
| 38 | 24 | 98 | 74 | 0.00250164 | 0.00220997 | 0.992 |
| 39 | 21 | 89 | 68 | 0.00277427 | 0.00222632 | 1.000 |
| 62 | 24 | 97 | 73 | 0.00250164 | 0.00220303 | 0.993 |
| 63 | 28 | 103 | 75 | 0.00227434 | 0.00218569 | 0.977 |
| 64 | 34 | 108 | 74 | 0.00208671 | 0.00216745 | 0.987 |

### 5.5 Reproducibility (flux scan)

- Case A: `python3 code/bimodality_flux_scan.py --sc-src 6 --aw-check`
- Case B: `python3 code/bimodality_flux_scan.py --sc-src 1 --aw-check`

## 6 Results: Case A (src = 6)

### 6.1 MC scan scope

- Parameter set: $N = 100,\ K = 6,\ n_0 = 1,\ \text{target} = 50,\ \text{src} = 6.$
- Number of dst values satisfying the bimodality criterion (Fig. 3 rule): 15, spanning 43–57.
- Special case: dst = target = 50 connects the shortcut directly to the absorber, yielding an early window and $P(C \geq 1) \approx 1$.

### 6.2 Analytic second-peak height vs. dst

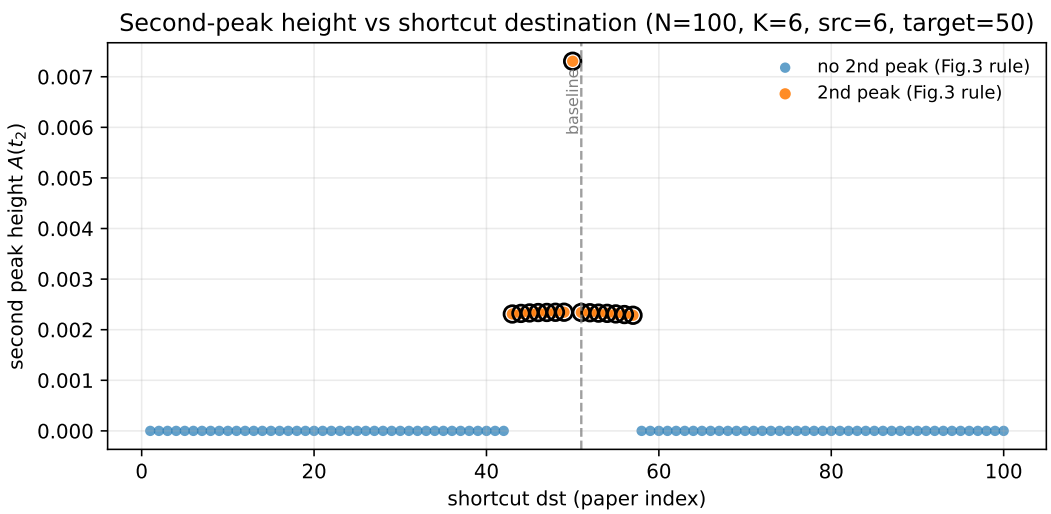


Figure 9: Analytic (Abate–Whitt inversion) second-peak height $A(t_2)$ as a function of shortcut destination dst.

### 6.3 Peak times (t1, t2) among bimodal cases

### 6.4 Mechanism: late second peak in Case A

- In Case A, src $\neq n_0$: a trajectory must first reach src = 6 before it can use the shortcut, so shortcut-induced effects typically manifest on later time scales.
- Bimodality can be interpreted as a mixture of two trajectory classes: those that reach src and utilize the shortcut relatively early (contributing to the earlier peak at $t_1$), and those that do not effectively utilize the shortcut until much later (contributing to the later peak at $t_2 \sim O(N)$).

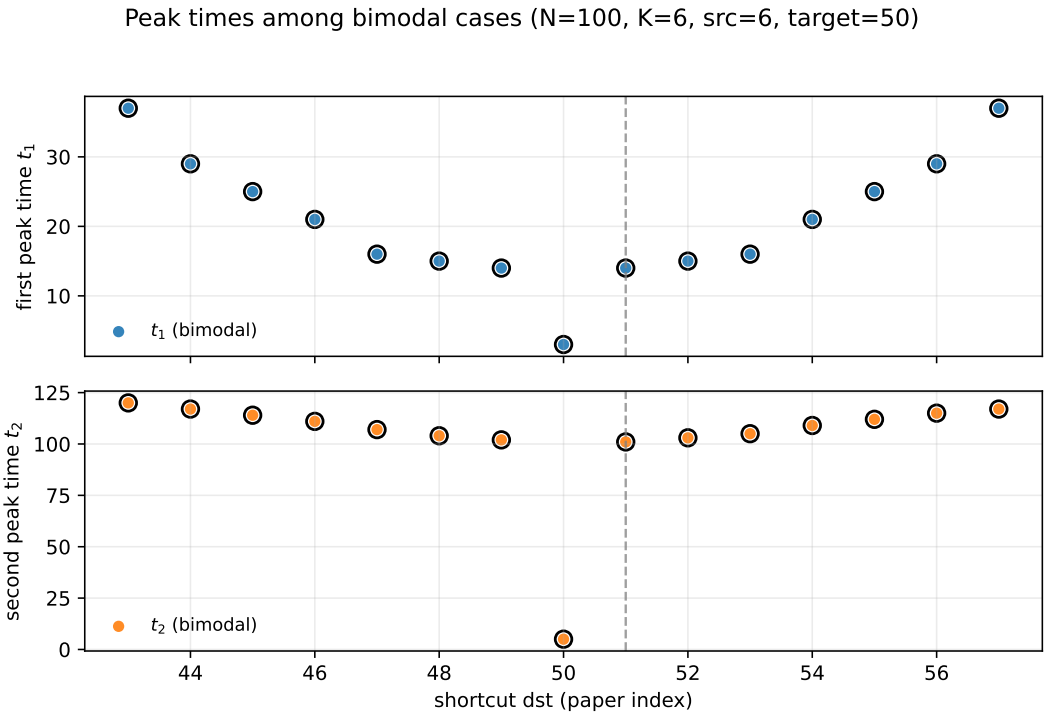


Figure 10: Peak locations $t_1$ and $t_2$ as functions of dst, restricted to cases satisfying the bimodality criterion.

- The special point dst = target connects directly to the absorbing state and produces an atypically early peak structure; it should be considered separately from typical late-second-peak behavior.

### 6.5 Representative FPT curves and second-peak windows

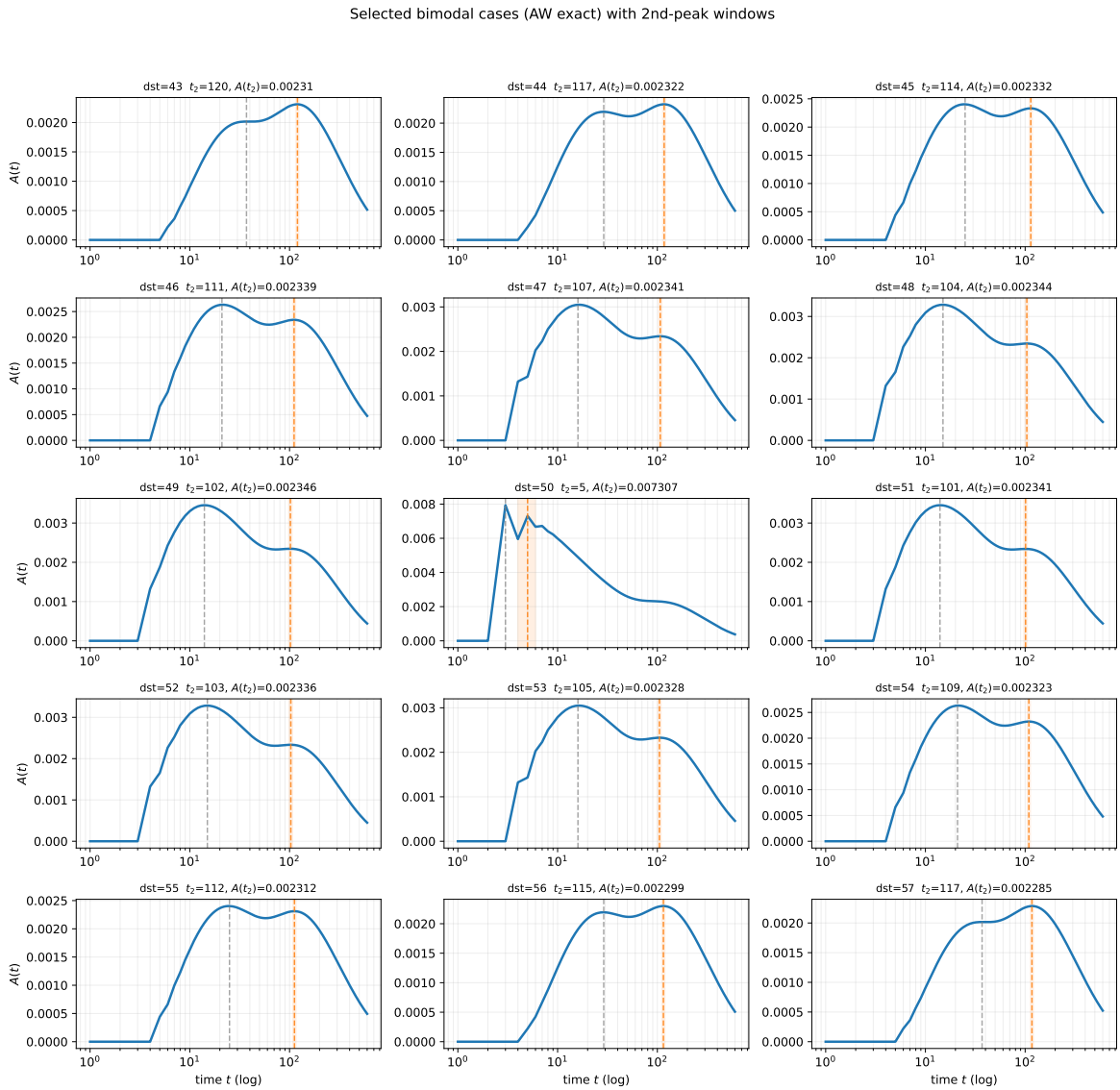


Figure 11: Analytic $A(t)$ (Abate–Whitt inversion) for all dst satisfying the bimodality criterion. Orange shading indicates the second-peak window $W = [t_2 - \Delta, t_2 + \Delta]$.

## 6.6 MC shortcut-crossing fractions within the second-peak window

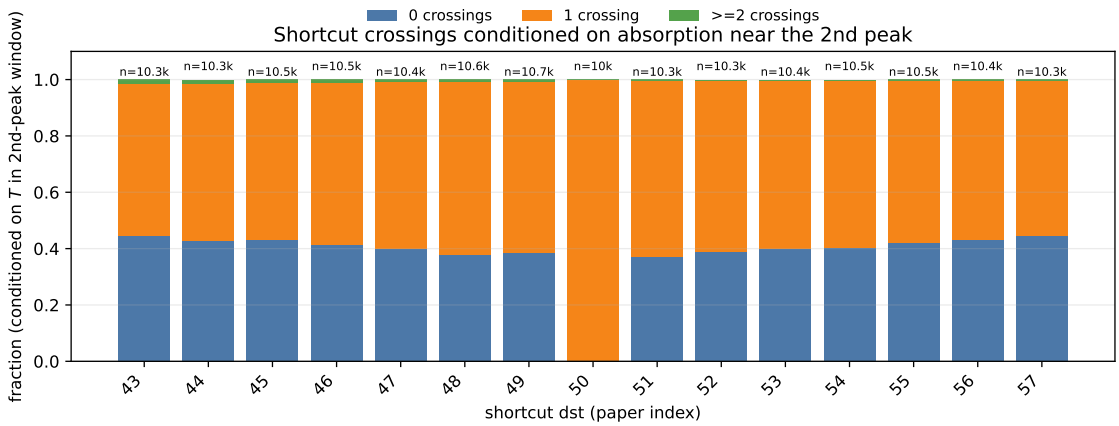


Figure 12: For each dst, conditional on $T \in W$, I estimate the fractions of trajectories with $C = 0$, $C = 1$, and $C \geq 2$ (stacked bars). The label atop each bar reports the window sample size $n$.

## 6.7 Relationship between second-peak prominence and shortcut usage

Define the peak-height ratio $R = A(t_2)/A(t_1)$. I examine how $P(C \geq 1 \mid T \in W)$ varies with $R$ and with the absolute height $A(t_2)$.

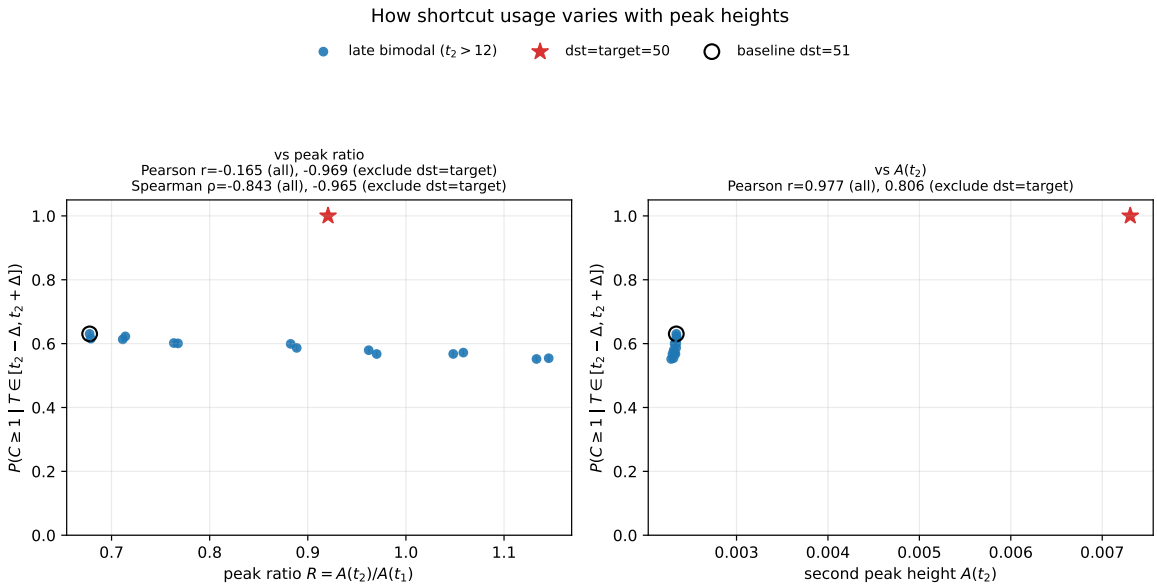


Figure 13: Left: $P(C \geq 1 \mid T \in W)$ vs. $R$; right: $P(C \geq 1 \mid T \in W)$ vs. $A(t_2)$. Error bars are Wilson 95% confidence intervals. Colors indicate early/late (early defined by $t_2 \leq 2K$; Case A is almost entirely late). The red star denotes the special case dst = target.

### 6.7.1 Correlation summary (computed by script)

- Pearson $r(R,\ P(C \geq 1))$: -0.165 (all), -0.969 (excluding dst = target).
- Spearman $\rho(R,\ P(C \geq 1))$: -0.843 (all), -0.965 (excluding dst = target).
- Pearson $r(A(t_2),\ P(C \geq 1))$: 0.977 (all), 0.806 (excluding dst = target).
- (Equivalently) Pearson $r(A(t_2),\ P(C = 0))$: -0.977 (all), -0.806 (excluding dst = target).

- (Late-only: $t_2 > 12$) Pearson $r(A(t_2),\ P(C \geq 1))$: 0.806; Pearson $r(R,\ P(C \geq 1))$: -0.969.

#### 6.7.2 Key takeaways (Case A)

- **Predominantly late bimodality:** aside from dst = target, bimodal cases in Case A are mainly late-second-peak instances (large $t_2$).
- **Nontrivial zero-crossing mass:** even conditional on $T \in W$, a substantial fraction of trajectories have $C = 0$, indicating the second peak reflects a slower subpopulation rather than paths that almost surely traverse the shortcut.
- **Monotone trend with $A(t_2)$:** larger $A(t_2)$ is associated with larger $P(C \geq 1 \mid T \in W)$ (equivalently, smaller $P(C = 0 \mid T \in W)$).
- **Special case** dst = target**:** the second peak occurs very early and $P(C \geq 1 \mid T \in W) \approx 1$, so this point should be discussed separately.

### 6.8 Summary table

| dst | $t_1$ | $t_2$ | $\Delta$ | $R$ | $A(t_1)$ | $A(t_2)$ | $P(C=0)$ | $P(C \geq 1)$ |
|---|---|---|---|---|---|---|---|---|
| 43 | 37 | 120 | 4 | 1.146 | 0.0020160646 | 0.00231041 | 0.446 | 0.554 |
| 44 | 29 | 117 | 4 | 1.059 | 0.0021926251 | 0.00232199 | 0.428 | 0.572 |
| 45 | 25 | 114 | 4 | 0.970 | 0.0024036907 | 0.00233158 | 0.432 | 0.568 |
| 46 | 21 | 111 | 4 | 0.889 | 0.0026312025 | 0.00233914 | 0.413 | 0.587 |
| 47 | 16 | 107 | 4 | 0.768 | 0.0030482422 | 0.00234085 | 0.399 | 0.601 |
| 48 | 15 | 104 | 4 | 0.714 | 0.0032836275 | 0.00234431 | 0.377 | 0.623 |
| 49 | 14 | 102 | 4 | 0.679 | 0.0034545803 | 0.00234556 | 0.384 | 0.616 |
| 50 | 3 | 5 | 1 | 0.921 | 0.0079321716 | 0.00730663 | 0.000 | 1.000 |
| 51 | 14 | 101 | 4 | 0.678 | 0.0034530973 | 0.0023412 | 0.369 | 0.631 |
| 52 | 15 | 103 | 4 | 0.711 | 0.0032842321 | 0.00233559 | 0.387 | 0.613 |
| 53 | 16 | 105 | 4 | 0.763 | 0.0030519913 | 0.00232787 | 0.398 | 0.602 |
| 54 | 21 | 109 | 4 | 0.883 | 0.0026314043 | 0.00232303 | 0.401 | 0.599 |
| 55 | 25 | 112 | 4 | 0.962 | 0.0024021309 | 0.00231205 | 0.421 | 0.579 |
| 56 | 29 | 115 | 4 | 1.048 | 0.0021939313 | 0.00229944 | 0.432 | 0.568 |
| 57 | 37 | 117 | 4 | 1.133 | 0.0020169903 | 0.00228525 | 0.448 | 0.552 |

## 7 Results: Case B (src = 1)

### 7.1 MC scan scope

- Parameter set: $N = 100,\ K = 6,\ n_0 = 1,\ \text{target} = 50,\ \text{src} = 1.$
- Number of dst values satisfying the bimodality criterion (Fig. 3 rule): 18, spanning 36–64.
- Special case: dst = target = 50 connects the shortcut directly to the absorber, yielding an early window and $P(C \geq 1) \approx 1$.

### 7.2 Analytic second-peak height vs. dst

### 7.3 Peak times (t1, t2) among bimodal cases

### 7.4 Why can "second peaks" occur at very early times in Case B?

- When src = $n_0$, the shortcut is available immediately: at step 1 the walker uses the shortcut $n_0 \to \text{dst}$ with probability $1/(K+1) = 1/7$.
- If dst is within one ring step of the target (i.e., dst $\in \{\text{target} \pm 1, \pm 2, \pm 3\}$ for $K = 6$), there exist extremely short absorbing paths

$$n_0 \xrightarrow{\text{shortcut}} \text{dst} \xrightarrow{\text{ring}} \text{target},$$

  which create a large absorption probability already at $t = 2$ and hence produce an early peak structure (and these early peaks are almost entirely composed of trajectories that traverse the shortcut).

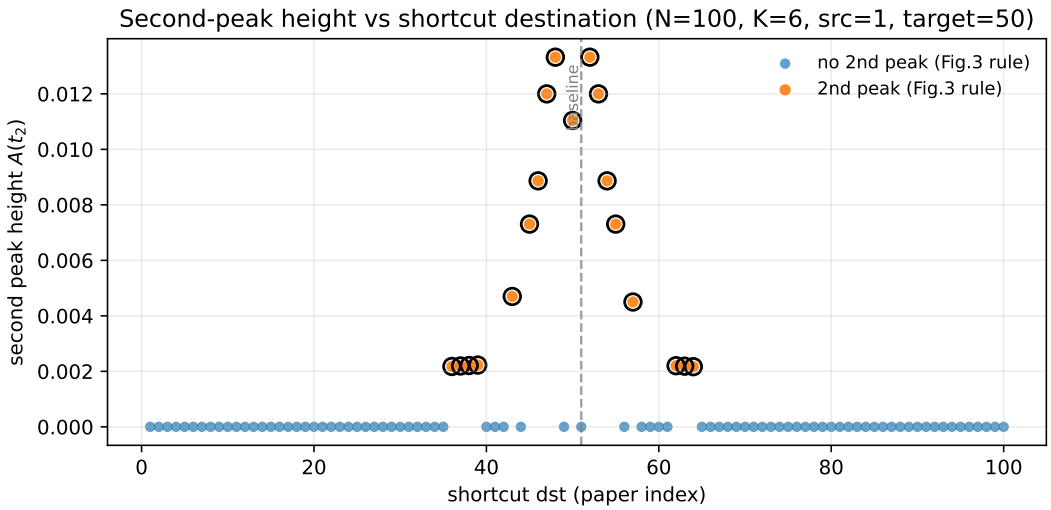


Figure 14: Analytic (Abate–Whitt inversion) second-peak height $A(t_2)$ as a function of shortcut destination dst.

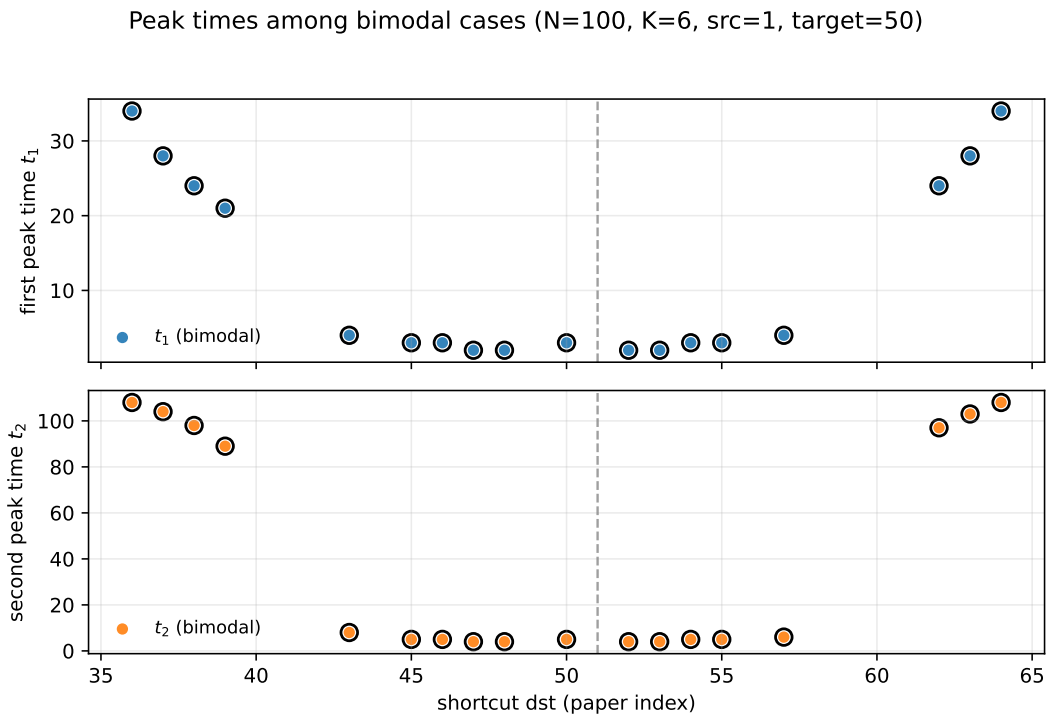


Figure 15: Peak locations $t_1$ and $t_2$ as functions of dst, restricted to cases satisfying the bimodality criterion.

- These early peaks are not numerical artifacts: for representative early-$t_2$ cases (e.g., dst = 47), comparison to exact master-equation propagation shows that Abate–Whitt inversion matches the exact values over the first 200 steps with maximum error $\sim 10^{-14}$ and agrees in both peak locations and heights.

### 7.5 Mechanism: late second peak in Case B

- Case B also exhibits late-second-peak instances: when the shortcut sends the walker far from the target, early shortcut usage can delay absorption.
- This separates trajectories into two classes: those that do not use the shortcut early (or are not substantially displaced by it), contributing to the earlier peak at $t_1$, and those that use the shortcut early and are pushed away from the target, contributing to the later peak at $t_2$.
- In these late-$t_2$ cases, conditional on $T \in W$ I typically observe $P(C \geq 1 \mid T \in W) \approx 0.63$–$0.72$, leaving a visible zero-crossing fraction.

### 7.6 Representative FPT curves and second-peak windows

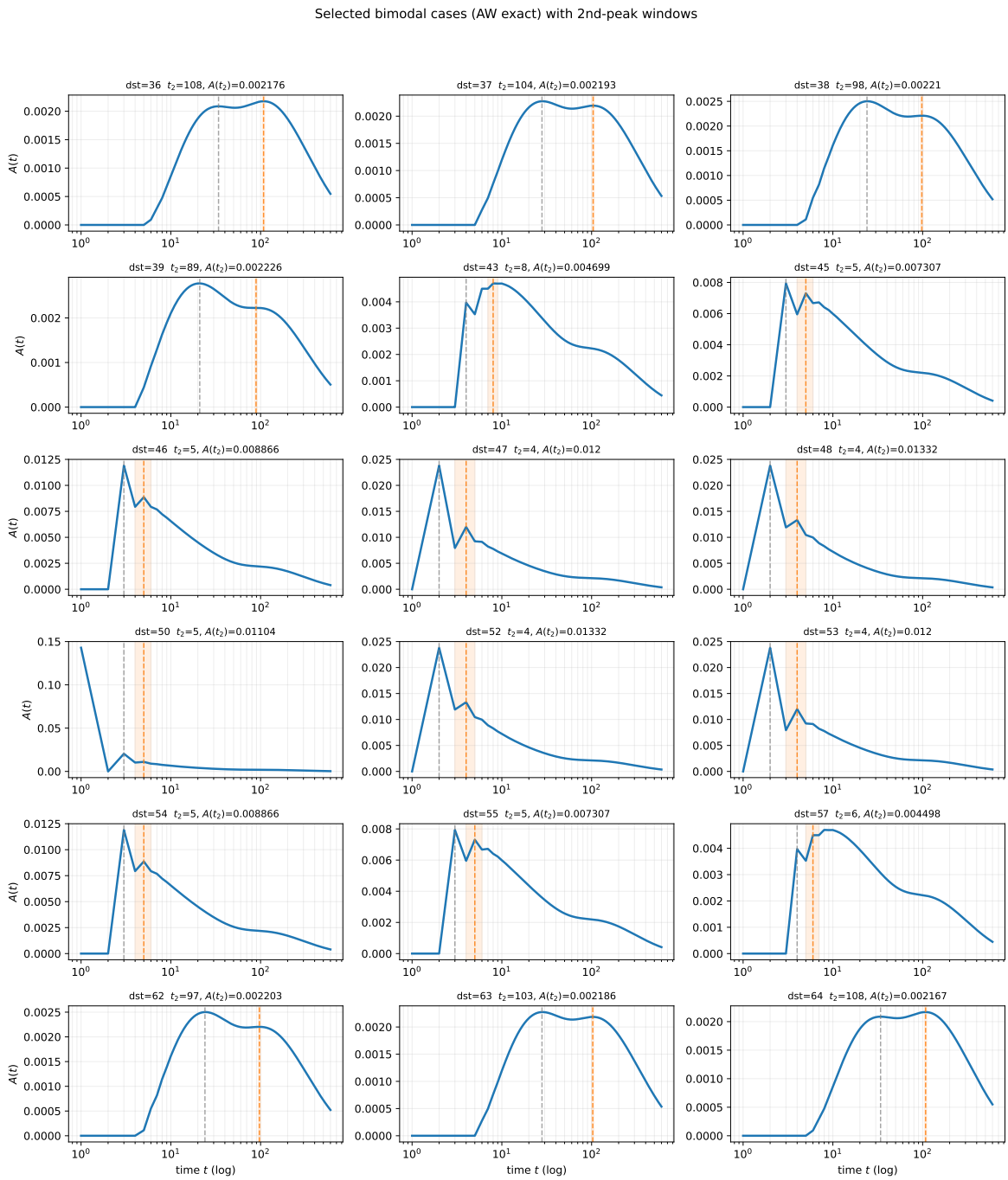


Figure 16: Analytic $A(t)$ (Abate–Whitt inversion) for all dst satisfying the bimodality criterion. Orange shading indicates the second-peak window $W = [t_2 - \Delta, t_2 + \Delta]$.

## 7.7 MC shortcut-crossing fractions within the second-peak window

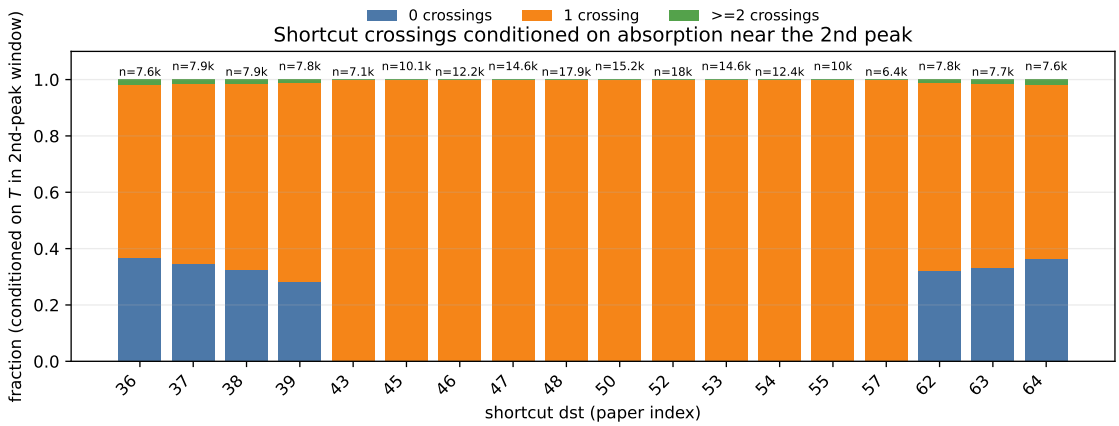


Figure 17: For each dst, conditional on $T \in W$, I estimate the fractions of trajectories with $C = 0$, $C = 1$, and $C \geq 2$ (stacked bars). The label atop each bar reports the window sample size $n$.

## 7.8 Relationship between second-peak prominence and shortcut usage

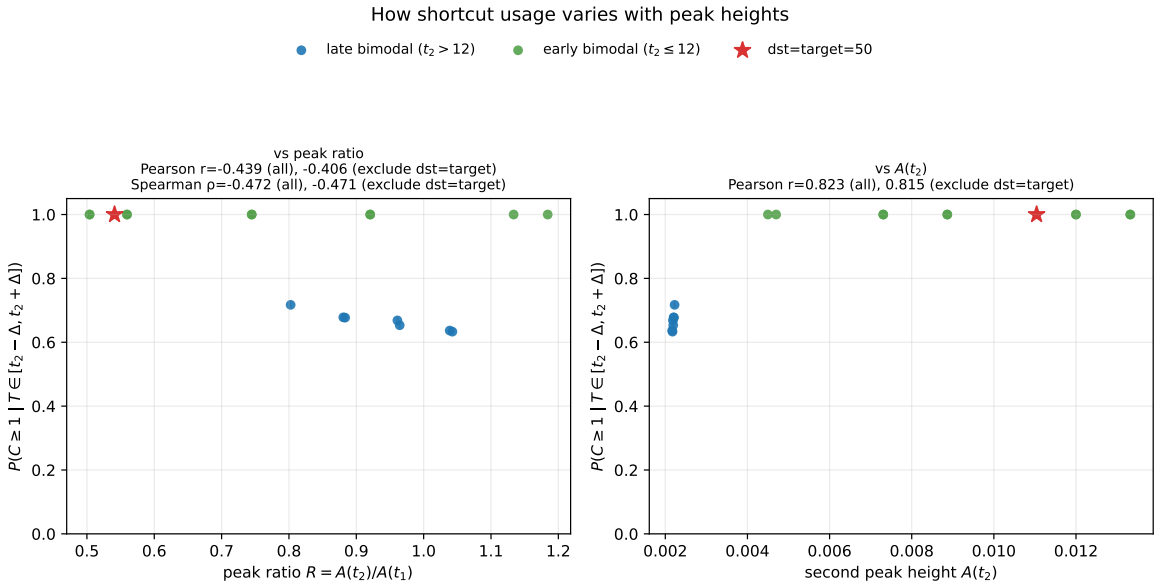


Figure 18: Left: $P(C \geq 1 \mid T \in W)$ vs. $R$; right: $P(C \geq 1 \mid T \in W)$ vs. $A(t_2)$. Error bars are Wilson 95% confidence intervals. Colors indicate early/late (early defined by $t_2 \leq 2K$). The red star denotes the special case dst = target.

### 7.8.1 Correlation summary (computed by script)

- Pearson $r(R,\ P(C \geq 1))$: -0.439 (all), -0.406 (excluding dst = target).
- Spearman $\rho(R,\ P(C \geq 1))$: -0.472 (all), -0.471 (excluding dst = target).
- Pearson $r(A(t_2),\ P(C \geq 1))$: 0.823 (all), 0.815 (excluding dst = target).
- (Equivalently) Pearson $r(A(t_2),\ P(C = 0))$: -0.823 (all), -0.815 (excluding dst = target).
- (Late-only: $t_2 > 12$) Pearson $r(A(t_2),\ P(C \geq 1))$: 0.938; Pearson $r(R,\ P(C \geq 1))$: -0.973.

### 7.8.2 Key takeaways (Case B)

- **Two mechanisms:** with src $= n_0$, bimodal cases typically split into an early-$t_2$ class (e.g., $t_2 \leq 8$) and a late-$t_2$ class (e.g., $t_2 \approx 90$–$110$).
- **Early-$t_2$ cases:** conditional on $T \in W$, trajectories almost surely traverse the shortcut, so $P(C \geq 1 \mid T \in W) \approx 1$ (and typically $C = 1$).
- **Late-$t_2$ cases:** conditional on $T \in W$, a substantial zero-crossing fraction remains (roughly 0.28–0.37 in this scan).
- **Trend with $A(t_2)$:** overall, smaller $A(t_2)$ is associated with smaller $P(C \geq 1 \mid T \in W)$ (equivalently, larger $P(C = 0 \mid T \in W)$). Thus the statement "a lower second peak implies a higher shortcut-usage fraction" is not supported here; the observed trend is the opposite.
- **Special case** dst = target**:** the shortcut points directly to the absorber, producing an early window and $P(C \geq 1 \mid T \in W) \approx 1$, and should be separated from the late-second-peak mechanism.

## 7.9 Summary table

| dst | $t_1$ | $t_2$ | $\Delta$ | $R$ | $A(t_1)$ | $A(t_2)$ | $P(C=0)$ | $P(C \geq 1)$ |
|---|---|---|---|---|---|---|---|---|
| 36 | 34 | 108 | 3 | 1.043 | 0.0020860594 | 0.00217576 | 0.367 | 0.633 |
| 37 | 28 | 104 | 3 | 0.964 | 0.0022753112 | 0.0021934 | 0.347 | 0.653 |
| 38 | 24 | 98 | 3 | 0.883 | 0.0025027973 | 0.00220997 | 0.323 | 0.677 |
| 39 | 21 | 89 | 3 | 0.802 | 0.0027759601 | 0.00222632 | 0.283 | 0.717 |
| 43 | 4 | 8 | 1 | 1.184 | 0.0039697399 | 0.00469917 | 0.000 | 1.000 |
| 45 | 3 | 5 | 1 | 0.921 | 0.0079321716 | 0.00730663 | 0.000 | 1.000 |
| 46 | 3 | 5 | 1 | 0.745 | 0.011900107 | 0.00886558 | 0.000 | 1.000 |
| 47 | 2 | 4 | 1 | 0.504 | 0.023807937 | 0.0119992 | 0.000 | 1.000 |
| 48 | 2 | 4 | 1 | 0.560 | 0.023789286 | 0.013322 | 0.000 | 1.000 |
| 50 | 3 | 5 | 1 | 0.541 | 0.020408317 | 0.0110409 | 0.000 | 1.000 |
| 52 | 2 | 4 | 1 | 0.560 | 0.023789286 | 0.013322 | 0.000 | 1.000 |
| 53 | 2 | 4 | 1 | 0.504 | 0.023807937 | 0.0119992 | 0.000 | 1.000 |
| 54 | 3 | 5 | 1 | 0.745 | 0.011900107 | 0.00886558 | 0.000 | 1.000 |
| 55 | 3 | 5 | 1 | 0.921 | 0.0079321716 | 0.00730663 | 0.000 | 1.000 |
| 57 | 4 | 6 | 1 | 1.134 | 0.003966843 | 0.0044984 | 0.000 | 1.000 |
| 62 | 24 | 97 | 3 | 0.881 | 0.0025011691 | 0.00220303 | 0.322 | 0.678 |
| 63 | 28 | 103 | 3 | 0.961 | 0.0022743913 | 0.00218569 | 0.332 | 0.668 |
| 64 | 34 | 108 | 3 | 1.039 | 0.0020860924 | 0.00216745 | 0.363 | 0.637 |

**Reproducibility:** flux scan (deterministic):

Case A: `python3 code/bimodality_flux_scan.py --sc-src 6 --aw-check`

Case B: `python3 code/bimodality_flux_scan.py --sc-src 1 --aw-check`

dst scan + MC:

Case A: `python3 code/dst_shortcut_usage_mc.py --sc-src 6 --mc-all-bimodal`

Case B: `python3 code/dst_shortcut_usage_mc.py --sc-src 1 --mc-all-bimodal`

Build: `latexmk -pdf -interaction=nonstopmode -halt-on-error`
`-auxdir=build -emulate-aux-dir valley_dst_report_en.tex`

(MC defaults to `--seed 0` unless specified).